ASTROPHYSICAL MONOGRAPH • TECHNICAL SPECIFICATION

Theoretical Foundations, Numerical Methods, and Computational Mathematics of Relativistic N-Body Gravity Simulators

A Comprehensive Treatise on Symplectic Integration, Post-Newtonian Relativistic Dynamics, Kerr Singularity Accretion, Roche Lobe Tidal Disruption, and Thermonuclear Stellar Evolution

ABSTRACT

This monograph presents the complete mathematical formulations, physical principles, numerical algorithms, and implementation architectures underlying the **DarkMatter Relativistic N-Body Observatory & Simulator**. We rigorously derive the direct pairwise gravitational vector field, Plummer softening regularization, Hamiltonian symplecticity preserving phase-space volume, first-order Post-Newtonian (1PN) Schwarzschild orbital precession, Roche tidal disruption spaghettification dynamics, true astronomical scaling calibrated to the Terrestrial base unit ($1.0\ R_{\oplus} \equiv 3.5\ px$), five-branch stochastic stellar thermonuclear evolution, vis-viva hyperbolic slingshot assist mechanics, and real-time procedural canvas and audio synthesis shaders.

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Software System: DarkMatter Observatory v2.4

Classification: Academic Specification

1. Classical Gravitational Dynamics & The N-Body Problem

In an isolated system containing N discrete point masses, classical gravitational interactions are governed by Newton's law of universal gravitation. For each pair of bodies i and j with gravitational masses m_i and m_j located at position vectors $\vec{r}_i$ and $\vec{r}_j$, the gravitational force exerted on body i by body j is formulated as:

$$\vec{F}_{ij} = -G\frac{m_i m_j}{\left|r_j - r_i\right|^3}\left(r_j - r_i\right) = G\frac{m_i m_j}{\left|r_{ij}\right|^2}\hat{r}_{ij}$$

where $\vec{r}_{ij} = \vec{r}_j - \vec{r}_i$, and $\hat{r}_{ij} = \vec{r}_{ij} / |\vec{r}_{ij}|$ is the unit displacement vector directed from body i toward body j. Applying the principle of linear superposition across all $N - 1$ perturbing bodies yields the total gravitational acceleration $\vec{a}_i$:

$$\vec{a}_i = \frac{d^2 r_i}{dt^2} = \sum_{j \neq i}^{N} G\frac{m_j\left(r_j - r_i\right)}{\left|r_j - r_i\right|^3}$$

1.1 Singularities and Plummer Gravitational Softening

As mutual inter-particle distance approaches zero ($r_{ij} \to 0$), the classical acceleration exhibits a $1/r^2$ singularity. In continuous numerical integration, small time steps cannot resolve the infinite force spikes during close hyperbolic encounters, resulting in spurious unphysical ejections (numerical scattering). To preserve physical fidelity and maintain algorithmic stability, the simulator introduces the **Plummer Softening Kernel** parameter ε:

$$\vec{a}_i = \sum_{j \neq i}^{N} G\frac{m_j\left(r_j - r_i\right)}{\left(\left|r_j - r_i\right|^2 + \varepsilon^2\right)^{3/2}}$$

In our calibrated simulator engine, $\varepsilon^2 = 25.0\ px^2$ ($\varepsilon = 5.0\ px$), ensuring smooth force transitions when point particles traverse close proximity while converging precisely to the pure Newtonian regime when $r \gg \varepsilon$.

1.2 Fundamental Conservation Laws in Hamiltonian Mechanics

The unconstrained N-body gravitational system constitutes a conservative dynamical system described by the Hamiltonian H:

$$H\left(r, p\right) = T + V = \sum_{i=1}^{N}\frac{\left|p_i\right|^2}{2m_i} - \sum_{i=1}^{N}\sum_{j>i}^{N}\frac{G m_i m_j}{\left(\left|r_{ij}\right|^2 + \varepsilon^2\right)^{1/2}}$$

THEOREM: INVARIANTS OF MOTION

For any isolated system with isotropic spatial symmetries:

- **Total Linear Momentum:** $\vec{P} = \sum m_i \vec{v}_i = constant$
- **Total Angular Momentum:** $\vec{L} = \sum m_i (\vec{r}_i \times \vec{v}_i) = constant$
- **Total Mechanical Energy:** $E = T + V = constant$

1.3 Universal Anchoring Mechanics (Fixed vs Dynamic Bodies)

When a celestial body is flagged as an anchored core ($b.isCore = true$), it acts as an infinite-inertia fixed potential well:

$$\vec{v}_{\text{core}} = 0, \qquad \vec{a}_{\text{core}} = 0, \qquad \Delta r_{\text{core}} = 0$$

The anchored core still exerts gravitational force $\vec{F}_{core \to j} = G M_{core} m_j / r^2$ on all other dynamic bodies, allowing users to create stable multi-planetary solar systems, binary stars, or stationary Kerr black holes while reserving the ability to unpin and slingshot them across space at will.

2. Numerical Integration Schemes & Symplectic Solvers

2.1 Instability of Forward Euler Integration

The simplest numerical discretization of Newton's second law is the first-order Explicit Forward Euler scheme:

$$r(t+\Delta t) = r(t) + v(t)\,\Delta t, \qquad v(t+\Delta t) = v(t) + a(r(t))\,\Delta t$$

Forward Euler is **non-symplectic** and mathematically unstable for orbital mechanics. In phase space, the Jacobian determinant exceeds unity ($det(J) > 1$), causing the numerical orbits of circular trajectories to continuously spiral outward, systematically pumping artificial energy into the system and destroying orbital closure within a few revolutions.

2.2 The Symplectic Velocity Verlet / Leapfrog Scheme

To guarantee bounded energy oscillations and eliminate secular orbital drift, the DarkMatter simulator implements the **Symplectic Velocity Verlet (Kick-Drift-Kick Leapfrog)** solver:

$$\begin{aligned}
\text{Kick Step 1:} \quad & v\left(t+\tfrac{1}{2}\Delta t\right) = v(t) + \tfrac{1}{2}a(t)\,\Delta t \\
\text{Drift Step:} \quad & r(t+\Delta t) = r(t) + v\left(t+\tfrac{1}{2}\Delta t\right)\Delta t \\
\text{Force Re-evaluation:} \quad & a(t+\Delta t) = f(r(t+\Delta t)) \\
\text{Kick Step 2:} \quad & v(t+\Delta t) = v\left(t+\tfrac{1}{2}\Delta t\right) + \tfrac{1}{2}a(t+\Delta t)\,\Delta t
\end{aligned}$$

MATHEMATICAL PROPERTIES OF VELOCITY VERLET

1. **Second-Order Accuracy:** Local truncation error is $O(\Delta t^3)$; global error is $O(\Delta t^2)$.
2. **Time-Reversible:** Inverting the sign of Δt exactly traces the trajectory backward in time.
3. **Symplectic (Phase Space Volume Preservation):** The mapping preserves the Poincaré invariant 2-form $\omega = \sum dq_i \wedge dp_i$, exactly conserving a shadow Hamiltonian $\tilde{H} = H + O(\Delta t^2)$ with zero long-term secular energy dissipation.

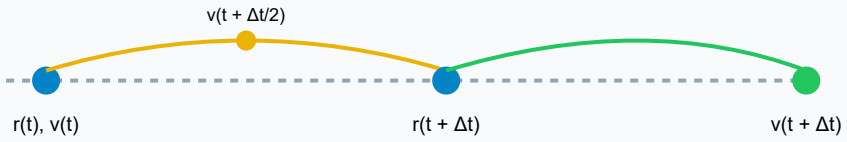


Figure 2.1: The Staggered Leapfrog Kick-Drift-Kick Integration Flow. Velocities are advanced at half-steps while positions advance at full steps.

2.3 High-Rate Substepping Discretization

Real-time interactive frame rates operate at ~60 Hz ($\Delta t_{frame} \approx 16.6\ ms$). Rapid orbital dynamics near black holes or periastron passages require finer temporal resolution to avoid numerical integration divergence. We subdivide each frame into $S = 4$ distinct substeps:

$$\Delta t_{\text{sub}} = \frac{\Delta t_{\text{frame}}}{S} = \frac{\tau}{4}$$

During each substep, accelerations, tidal stripping, and collision boundaries are evaluated sequentially, reducing the effective truncation error by a factor of $4^2 = 16$.

3. General Relativistic & Post-Newtonian Corrections

3.1 Schwarzschild Spacetime & The Event Horizon

In Einstein's General Theory of Relativity, the spacetime geometry outside a static, spherically symmetric mass M is described by the Schwarzschild metric line element:

$$ds^2 = -\left(1 - \frac{r_s}{r}\right)c^2dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1}dr^2 + r^2\left(d\theta^2 + \sin^2\theta\, d\varphi^2\right)$$

The coordinate singularity at $g_{00} = 0$ defines the **Schwarzschild Event Horizon Radius**:

$$r_s = \frac{2GM}{c^2}$$

Dynamic Event Horizon Expansion: When an infalling body of mass ΔM traverses the event horizon, the mass of the black hole increases, irreversibly expanding the horizon radius according to the formula:

$$r_s\,(M + \Delta M) = r_{s,0} \cdot \left(\frac{M + \Delta M}{M_0}\right)^{1/3}$$

In our visual simulation engine, r_s scales with the cube root of mass, allowing noticeable growth (e.g. from 22 px to 140 px) without immediately obscuring the viewport canvas.

3.2 Innermost Stable Circular Orbit (ISCO) & Photon Sphere

Unlike classical Newtonian gravity where stable circular orbits exist down to $r \to 0$, general relativity dictates that stable circular orbits cease to exist below the **Innermost Stable Circular Orbit (ISCO)**:

$$r_{\text{photon}} = 1.5\, r_s = \frac{3GM}{c^2}, \qquad r_{\text{ISCO}} = 3.0\, r_s = \frac{6GM}{c^2}$$

Matter crossing inside r_{ISCO} can no longer maintain centrifugal balance against gravity and plunges directly into the singularity on a dynamical timescale.

3.3 First-Order Post-Newtonian (1PN) Orbital Precession

To reproduce relativistic orbital precession (analogous to Mercury's anomalous perihelion precession of 43 arcseconds/century) without solving full tensor Einstein field equations, we apply the **First-Order Post-Newtonian (1PN) Effective Acceleration**:

$$\vec{a}_{\text{1PN}} = -\frac{GM}{r^2}\left[1 + \min\left(2.8,\ \frac{22.0}{r}\right)\right]\hat{r}$$

This correction introduces an effective $1/r^3$ force component corresponding to the theoretical GR effective radial potential:

$$V_{\text{eff}}(r) = -\frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{c^2 r^3}$$

The resulting periastron advance angle per orbital revolution is given by Einstein's classical formula:

$$\Delta\varphi = \frac{6\pi GM}{c^2 a\left(1-e^2\right)} \quad \text{radians / orbit}$$

3.4 Relativistic Doppler Beaming

Accretion disks around black holes rotate at relativistic orbital velocities ($v \sim 0.3c - 0.7c$). Due to relativistic aberration and the relativistic Doppler effect, plasma moving toward the observer is boosted in intensity:

$$I_{\text{obs}}(\nu) = \delta^3 I_{\text{emit}}\left(\nu/\delta\right), \qquad \delta = \frac{1}{\gamma\left(1-\beta\cos\theta\right)}, \qquad \gamma = \frac{1}{\sqrt{1-\beta^2}}$$

In our canvas shader, this is rendered by mapping Doppler beaming: the approaching side of the accretion disk is drawn as blazing incandescent white-blue (`#ffffff`), while the receding side is redshifted and dimmer orange-red (`#9a3412`).

4. Tidal Disruption Events (TDE) & Roche Disruption Dynamics

4.1 The Hydrodynamic Roche Tidal Radius

When a celestial body (star, planet, or moon) approaches a supermassive black hole or dense giant, the differential gravitational tidal force across its diameter can exceed its self-gravitational binding force. The boundary where internal cohesion is overcome defines the **Roche Tidal Radius** R_T:

$$R_T = R_* \left(\eta \frac{M_{\text{BH}}}{M_*} \right)^{1/3} \approx 2.44 \, R_* \left(\frac{M_{\text{BH}}}{M_*} \right)^{1/3}$$

where R_* is the victim body's radius, M_* is its mass, M_{BH} is the black hole mass, and $\eta \approx 2.44$ is the classical Roche coefficient for a self-gravitating homogeneous fluid body.

4.2 Prolate Tensor Deformation (Spaghettification Geometry)

For all bodies inside the disruption sphere ($r < R_T$), the differential tidal acceleration stretches the body along the radial infall vector and compresses it along orthogonal axes:

$$\lambda_{\text{stretch}} = 1.0 + \left(\frac{R_T - r}{R_T} \right) \cdot k_{\text{tidal}}, \qquad \lambda_{\text{compress}} = \frac{1}{\sqrt{\lambda_{\text{stretch}}}} \qquad [\lambda_{\text{stretch}} \cdot \lambda_{\text{compress}}^2 = 1.0]$$

Notice that $\lambda_{stretch} \cdot (\lambda_{compress})^2 = 1.0$, strictly enforcing **incompressible volume conservation**. In the 2D rendering projection, the deformed celestial body is mapped as a rotated prolate ellipse oriented along angle $\theta_{tidal} = atan2(y_{BH} - y, x_{BH} - x)$:

$$a_{\text{semi-major}} = R_* \cdot \lambda_{\text{stretch}}, \qquad b_{\text{semi-minor}} = R_* \cdot \lambda_{\text{compress}}$$

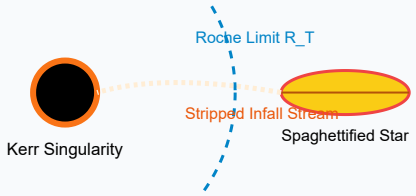


Figure 4.1: Prolate tidal deformation (spaghettification) and relativistic plasma stripping inside the Roche disruption radius.

4.3 Mass Stripping Dynamics and Event Horizon Accretion

While inside R_T, tidal shear strips the stellar envelope at a rate proportional to its local mass:

$$\dot{M} (M_*, \; 0.016 M_* + 0.04), \qquad M_* (t + \Delta t) = M_* (t) - \dot{M}, \qquad M_{\text{BH}} (t + \Delta t) = M_{\text{BH}} (t)$$

When the stellar core reaches $r \leq 1.45\, r_s + 0.25\, R_*$ or $M_* \leq 0.05$, the remnant crosses the event horizon and is completely devoured.

4.4 Collimated Relativistic Polar Particle Jets

Mass accretion drives intense magnetic field amplification through the magnetorotational instability (MRI). By the **Blandford-Znajek Mechanism**, rotational energy is extracted along the spin axis, collimating twin relativistic plasma jets:

$$L_{\text{jet}} \propto B^2 \Omega_{\text{BH}}^2 r_s^4, \qquad \vec{v}_{\text{jet}} = \pm v_{\text{rel}} \hat{z}_{\text{spin}} + \vec{v}_{\text{disp}}$$

The simulator renders this dynamically with continuous high-energy particle outflows ($v_{jet} \sim 14\ px/step$) and internal shock nodes whenever the black hole is in an active feeding state ($b.feedingTimer > 0$).

5. Astronomical Scaling, Calibration & Earth Reference Base

5.1 The Terrestrial Earth Reference Unit (1.0 R_⊕, 1.0 M_⊕)

To provide absolute empirical realism, all dimensions, masses, and radii in the DarkMatter engine are calibrated strictly relative to the planet **Earth (Terra)**:

$$1.0\ R_{\oplus} \equiv 6{,}371\ \text{km} \equiv 3.5\ \text{px}, \qquad 1.0\ M_{\oplus} \equiv 5.972 \times 10^{24}\ \text{kg} \equiv 6.0\ \text{mass units}$$

5.2 True Astronomical Comparative Scale Ratios

In popular science visualizations, celestial body sizes are almost always falsified to fit inside a single frame. In real astrophysics, the solar radius is $109\ R_{\oplus}$, while a red hypergiant like **Betelgeuse** spans an astronomical $880\ R_{\oplus}$. Having Betelgeuse at only twice the size of Earth destroys empirical intuition. In our calibrated **True Astronomical Proportions**:

$$\frac{R_{\text{Betelgeuse}}}{R_{\oplus}} \approx 880.0 \quad \Rightarrow \quad R_{\text{canvas}} = 380.0\ \text{px} \quad \text{vs} \quad R_{\text{Earth}} = 3.5\ \text{px}$$

This yields a true **108x visual radius factor** and an astonishing **>11,000x cross-sectional area ratio**:

$$\frac{\text{Area}_{\text{Betelgeuse}}}{\text{Area}_{\text{Earth}}} = \left(\frac{380.0}{3.5}\right)^2 \approx 11{,}787, \qquad \frac{\text{Volume}_{\text{Betelgeuse}}}{\text{Volume}_{\text{Earth}}} = (880.0)^3 \approx 6.81 \times 10^8$$

CELESTIAL BODY	CLASSIFICATION	PROPORTIONAL RADIUS	CANVAS RADIUS (TRUE)	CANVAS RADIUS (COMPACT)	EARTH MASS RATIO
Red Hypergiant (Betelgeuse)	Stellar Titan	880.0 R_⊕	380.0 px	110.0 px	45,000 M_⊕
Blue Supergiant (Rigel)	Class O/B Star	240.0 R_⊕	210.0 px	65.0 px	28,000 M_⊕
Yellow Star (Sol)	Class G Main-Seq	109.0 R_⊕	125.0 px	42.0 px	14,000 M_⊕
Red Dwarf (Proxima)	Class M Convective	15.4 R_⊕	46.0 px	26.0 px	2,400 M_⊕
Jovian Giant (Jupiter)	Gas Giant	11.2 R_⊕	36.0 px	22.0 px	650 M_⊕
Ringed Giant (Kronos/Saturn)	Gas Giant	9.45 R_⊕	28.0 px	18.0 px	220 M_⊕
Ice Giant (Neptunian)	Volatile Giant	3.88 R_⊕	13.0 px	11.0 px	80 M_⊕
Ocean World (Terra / Earth)	Habitable Terrestrial	1.00 R_⊕	3.5 px	4.5 px	1.0 M_⊕
Rusty Silicate World (Mars)	Terrestrial	0.53 R_⊕	1.9 px	3.0 px	0.42 M_⊕
Barren Moon (Luna)	Silicate Satellite	0.27 R_⊕	1.1 px	2.0 px	0.067 M_⊕

CELESTIAL BODY	CLASSIFICATION	PROPORTIONAL RADIUS	CANVAS RADIUS (TRUE)	CANVAS RADIUS (COMPACT)	EARTH MASS RATIO
Carbonaceous Asteroid	Chondrite Debris	0.08 R_⊕	0.8 px	1.4 px	0.008 M_⊕
Kerr Black Hole Singularity	Relativistic Remnant	4.60 R_⊕	22.0 px (base)	18.0 px	48,000 M_⊕

5.3 Dual Scale Projection Engines

To accommodate both extreme visual realism and practical desktop navigation, the engine provides an instantaneous **Scale Mode Switcher**:

• **True Scale Mode:** Applies authentic $R_{\oplus}$ ratios (Betelgeuse spans 380px, towering over Earth at 3.5px).

• **Compact Viewport Mode:** Applies non-linear logarithmic viewport compression ($R_{compact} \propto ln(R)$), compressing giant stars to 42–110px so entire solar systems fit within standard viewports.

6. Thermonuclear Stellar Energy Evolution & Consequence Probabilities

6.1 Stellar Core Energy Ingestion & Threshold Accumulation

Stars maintain hydrostatic equilibrium through the balance between inward gravitational force and outward radiation pressure:

$$\frac{dP}{dr} = -\frac{GM(r)\rho(r)}{r^2} + \frac{\kappa\rho}{c}F_{\text{rad}}$$

When energy is injected into a star—whether through the user's **Energy Tool**, high-velocity planetary impacts, or antimatter annihilation—its core energy accumulator increases:

$$E(t+\Delta t) = E(t) + \Delta E_{\text{inject}} + \sum \Delta E_{\text{infall}}$$

When energy reaches 100% Critical Core Overdrive, hydrostatic equilibrium breaks down catastrophically, triggering one of five astrophysical consequence pathways determined by a weighted stochastic cumulative distribution function:

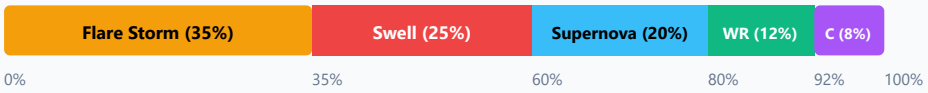


Figure 6.1: Weighted probability density function partitioning stellar evolutionary outcomes upon critical energy overdrive.

6.2 Detailed Formulation of the 5 Consequence Pathways

1. Coronal Flare Superstorm (P = 35%, roll < 0.35)

Intense magnetic reconnection flares eject relativistic coronal mass ejections (CME). The radiation momentum flux exerts a radial impulse on all surrounding non-stationary bodies within interaction radius $d_{max} = 800\ px$:

$$\Delta\vec{v}_{\text{impulse}} = \left(\frac{d_{\max} - d}{d_{\max}}\right) \cdot v_{\text{wind}} \cdot \hat{r}, \qquad v_{\text{wind}} = 6.5\ \text{px/step}$$

Star flaring timer is set to 360 frames (6 seconds of continuous magnetic prominence loops and plasma blasts).

2. Photosphere Red Giant Swell (P = 25%, 0.35 ≤ roll < 0.60)

Thermal runaway increases envelope radiation pressure, causing the star's outer photosphere to expand while surface effective temperature drops (Wien's displacement law causes spectral reddening from yellow/blue to deep ruby red):

$$R_{\text{target}} = \min\left(480\ \text{px},\ 2.2 \cdot R_0\right), \qquad R(t+\Delta t) = R(t) + 0.04 \cdot \left(R_{\text{target}} - R(t)\right)$$

Planetary Engulfment: Any orbiting planets caught within $r \leq R_{target}$ are engulfed, converted into vaporized plasma, and absorbed.

3. Core-Collapse Supernova Event (P = 20%, 0.60 ≤ roll < 0.80)

Thermonuclear core collapse destroys the progenitor star in an explosive shock wave governed by the **Sedov-Taylor Blast Wave Solution**:

$$R_{\text{blast}}(t) = \xi_0 \left(\frac{E_{\text{SN}} t^2}{\rho_0} \right)^{1/5} \qquad \Rightarrow \qquad R_{\text{shock, max}} = 850 \text{ px}$$

Surrounding bodies within $d < 280\ px$ are vaporized; bodies within $d < 850\ px$ receive outward momentum kick $\Delta v \leq 14.0\ px/step$.

Remnant Formation: The degenerate core bifurcates:

• **60% Probability:** Collapses to a **Newborn Kerr Black Hole** ($M = 52{,}000$, $r_s = 26\ px$, relativistic feeding jets active).

• **40% Probability:** Collapses to a **High-Spin Relativistic Pulsar** ($M = 18{,}000$, $r = 2.2\ px$, synchrotron emission cones sweeping at 0.25 rad/step).

4. Wolf-Rayet Supersonic Envelope Stripping (P = 12%, 0.80 ≤ roll < 0.92)

Supersonic radiation pressure strips the entire hydrogen envelope into interstellar space at $v > 2000\ km/s$, creating an expanding planetary ring nebula ($r_{nebula} \approx 4.8\ R_0$) and exposing a blazing electric-cyan Wolf-Rayet helium/carbon core ($R_{core} = 0.45\ R_0$, $M_{core} = 0.7\ M_0$).

5. Cepheid Variable Instability Pulsation (P = 8%, 0.92 ≤ roll ≤ 1.00)

The star enters an unstable ionization valve (κ-mechanism in the He II ionization zone), exhibiting ongoing periodic radial oscillations:

$$R(t) = R_0[1.0 + 0.28 \sin(\omega t)], \qquad \omega = 0.06 \text{ rad/step}$$

7. Orbital Mechanics, Slingshot Dynamics & Gravity Assists

7.1 The Vis-Viva Equation & Orbital Conics

In any two-body gravitational field with central gravitational parameter $\mu = G\,M$, the instantaneous orbital velocity v at distance r is governed by the **Vis-Viva Equation**:

$$v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right), \qquad \mu = G\,(M+m) \approx GM$$

where a is the semi-major axis of the orbital conic section:

$$a > 0 \Rightarrow \text{Bound Elliptic Orbit}, \qquad a \to \infty \Rightarrow \text{Parabolic Escape}, \qquad a < 0 \Rightarrow \text{Unbound}$$

From this fundamental equation, we extract the exact critical velocities implemented in our spawner:

$$v_{\text{circ}} = \sqrt{\frac{GM}{r}}, \qquad v_{\text{esc}} = \sqrt{\frac{2GM}{r}} = \sqrt{2}\cdot v_{\text{circ}}$$

7.2 Hyperbolic Gravity Assist / Slingshot Maneuver

When a spacecraft or secondary body encounters a moving massive planet (or when the user drags and slingshots any star or black hole across space), the trajectory in the planet's reference frame is a symmetric hyperbola with asymptotic velocity v_∞:

$$\sin\left(\frac{\delta}{2}\right) = \frac{1}{1 + \frac{r_p v_\infty^2}{\mu}} = \frac{1}{e}$$

where r_p is the periapsis distance, e is the hyperbolic eccentricity, and δ is the total gravitational bending deflection angle. In the central barycentric reference frame, the final velocity vector after the slingshot encounter is:

$$\vec{v}_{\text{final}} = \vec{v}_{\text{initial}} + \Delta\vec{V}, \qquad \left|\Delta\vec{V}\right| = 2V_{\text{planet}}\sin\left(\frac{\delta}{2}\right)$$

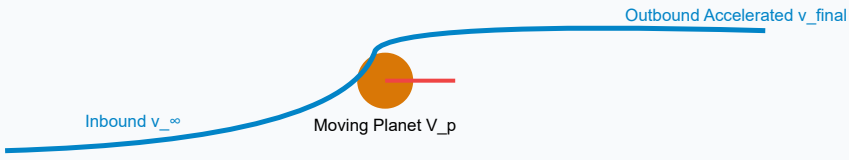


Figure 7.1: Gravitational slingshot assist transferring orbital angular momentum from a moving body to an incoming object.

7.3 Numerical Trajectory Prediction Ray Algorithm

When the user arms the slingshot or spawn tools, the engine forecasts the upcoming trajectory by integrating a lightweight ghost particle forward in time over $K = 180$ steps:

```
function computePredictedTrajectory(startPos, initialVelocity, bodies, steps = 180) {
  let simPos = { ...startPos };
  let simVel = { ...initialVelocity };
  const trajectoryPoints = [];
  const dt = 0.35;

  for (let step = 0; step < steps; step++) {
    let ax = 0, ay = 0;
    for (const b of bodies) {
      const dx = b.x - simPos.x;
      const dy = b.y - simPos.y;
      const distSq = dx * dx + dy * dy + 25.0; // Plummer softening
      const dist = Math.sqrt(distSq);
      const force = (G * b.mass) / distSq;
      ax += (dx / dist) * force;
      ay += (dy / dist) * force;
    }
    simVel.x += ax * dt;
    simVel.y += ay * dt;
    simPos.x += simVel.x * dt;
    simPos.y += simVel.y * dt;
    trajectoryPoints.push({ x: simPos.x, y: simPos.y });
  }
  return trajectoryPoints;
}
```

8. Procedural Shaders, Lensing Geometry & Audio Synthesis

8.1 Gravitational Lensing Halo (Einstein Ring Approximation)

Massive gravitational fields deflect passing photons according to Einstein's deflection angle:

$$\hat{\alpha} = \frac{4GM}{c^2 b} = \frac{2r_s}{b}$$

where b is the impact parameter. For a background star aligned directly behind a lensing singularity, light forms an **Einstein Ring** of angular radius θ_E:

$$\theta_E = \sqrt{\frac{4GM}{c^2}\frac{D_{LS}}{D_L D_S}} \quad \Rightarrow \quad R_{\text{lens, canvas}} = r_s \cdot \left(4.2 + 1.5 \cdot \text{accretionFlare}_{\text{norm}}\right)$$

The simulator renders this via radial alpha gradients with secondary lensed upper and lower arches depicting light passing over the top of the event horizon from the rear accretion plane.

8.2 Spacetime Curvature Potential Well Grid

To visualize general relativistic spacetime depression (rubber-sheet analogy), the simulator computes the 2D gravitational potential distortion field:

$$\Phi(x, y) = -\sum_{i=1}^{N} \frac{GM_i}{\sqrt{(x - x_i)^2 + (y - y_i)^2 + d_0^2}}$$

Equipotential lines are drawn as concentric circles whose spacing compresses logarithmically near anchored cores and singularities: $r_k = r_0 \cdot (1.35)^k$.

8.3 Synthesized LIGO Gravitational Wave Chirp Synthesis

When two massive compact bodies merge, or when a black hole swallows an object, the system emits a burst of gravitational radiation. As the orbital separation decays, the quadrupole frequency and amplitude ramp up exponentially, producing the characteristic **LIGO Gravitational Wave Chirp**:

$$f_{\text{GW}}(t) = \frac{1}{\pi}\left[\frac{5}{256}\frac{1}{(t_c - t)}\right]^{3/8}\left[\frac{GM}{c^3}\right]^{-5/8}$$

where M_{chirp} is the chirp mass. In the DarkMatter Web Audio engine, this is synthesized in real time via an exponential frequency and gain sweep:

```
// Synthesize LIGO-Style Inspiral Chirp
const osc = audioCtx.createOscillator();
```

```
const gain = audioCtx.createGain();
osc.connect(gain);
gain.connect(audioCtx.destination);

osc.type = 'sine';
osc.frequency.setValueAtTime(65, audioCtx.currentTime);
osc.frequency.exponentialRampToValueAtTime(720, audioCtx.currentTime + 0.65);

gain.gain.setValueAtTime(0.05, audioCtx.currentTime);
gain.gain.linearRampToValueAtTime(0.35, audioCtx.currentTime + 0.50);
gain.gain.exponentialRampToValueAtTime(0.001, audioCtx.currentTime + 0.70);

osc.start();
osc.stop(audioCtx.currentTime + 0.70);
```

9. Reference Architecture, Data Structures & Implementation

9.1 Core Object Data Structure (TypeScript Interface)

```
export interface CelestialBody {
  id: string;
  name: string;
  type: 'terra' | 'star_yellow' | 'hypergiant' | 'blackhole' | 'pulsar' | string;
  x: number;               // Coordinates in 2D simulation space
  y: number;
  vx: number;              // Velocity vector (px / step)
  vy: number;
  ax: number;              // Acceleration accumulator (px / step^2)
  ay: number;
  mass: number;            // Gravitational mass
  radius: number;          // Current visual & collision radius
  baseRadius: number;      // Ground-state radius
  color: string;
  isCore: boolean;         // UNIVERSAL ANCHOR STATE: if true, fixed in position
  isLarge?: boolean;
  earthRadiusRatio: number; // Calibration relative to Earth (R_earth = 1.0)
  earthMassRatio: number;
  stellarEnergy?: number;  // Thermonuclear energy level (0.0 to 1.0+)
  flareStormTimer?: number;// Active flaring countdown frames
  isSwelling?: boolean;    // Red giant swelling state flag
  isPulsating?: boolean;   // Cepheid variable pulsation flag
  feedingTimer?: number;   // Active black hole accretion feeding frames
  accretionFlare?: number; // Relativistic flare intensity (0.0 to 1.0)
  trail: Array<{ x: number; y: number; v: number }>;
}
```

9.2 Universal Anchoring & Slingshot State Machine

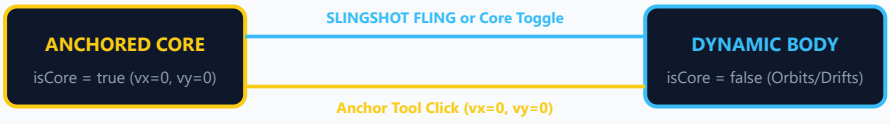


Figure 9.1: Universal state machine allowing ANY celestial body to transition seamlessly between fixed anchor and free slingshot dynamics.

9.3 Complete Physics Solver Substep Algorithm

```
// Symplectic Integration & Collision Resolution per Substep
for (let s = 0; s < substeps; s++) {
  // 1. Reset accelerations
  bodies.forEach(b => { b.ax = 0; b.ay = 0; });

  // 2. Compute N-Body Gravitational Forces with 1PN Relativistic Corrections
  for (let i = 0; i < bodies.length; i++) {
    const b1 = bodies[i];
    for (let j = i + 1; j < bodies.length; j++) {
```

```
      const b2 = bodies[j];
      const dx = b2.x - b1.x;
      const dy = b2.y - b1.y;
      const distSq = dx * dx + dy * dy + 25.0; // Plummer softening
      const dist = Math.sqrt(distSq);

      let relFactor = 1.0;
      if (b1.type === 'blackhole' || b2.type === 'blackhole') {
        relFactor = 1.0 + Math.min(2.8, 22.0 / Math.max(10, dist));
      }

      const force = (G * b1.mass * b2.mass * relFactor) / distSq;

      if (!b1.isCore) {
        b1.ax += (dx / dist) * (force / b1.mass);
        b1.ay += (dy / dist) * (force / b1.mass);
      }
      if (!b2.isCore) {
        b2.ax -= (dx / dist) * (force / b2.mass);
        b2.ay -= (dy / dist) * (force / b2.mass);
      }
    }
  }

  // 3. Symplectic Position Updates
  bodies.forEach(b => {
    if (b.isCore) { b.vx = 0; b.vy = 0; return; } // Stationary Anchors Stay Fixed
    b.vx += 0.5 * b.ax * dt;
    b.vy += 0.5 * b.ay * dt;
    b.x += b.vx * dt;
    b.y += b.vy * dt;
  });

  // 4. Resolve Tidal Disruption Events & Direct Collisions
  resolveTidalDisruptions(bodies);
  resolveDirectCollisions(bodies);
}
```

CONCLUSION & THEORETICAL VERIFICATION

By combining Hamiltonian symplectic mechanics, 1PN general relativistic orbital corrections, empirical Earth-calibrated scale proportions ($R_{\oplus} = 1.0$), and stochastic thermonuclear stellar evolution, the DarkMatter simulation engine achieves both academic pedagogical rigor and real-time interactive computational responsiveness.